

ICS 6B: Homework 2

1.7.3

(a)

$$\exists x \neg B(x)$$

(b)

$$\forall x B(x)$$

(c)

$$T(\text{Sam}) \wedge \neg B(\text{Sam})$$

(d)

$$\exists x (\neg T(x) \wedge B(x))$$

(e)

$$\forall x (T(x) \rightarrow B(x))$$

1.7.7

(b)

$$\forall x((x \neq \text{Nima}) \rightarrow N(x))$$

Everyone other than Nima is a new employee.

This is **True**

(c)

$$\exists x(\neg D(x) \wedge N(x))$$

Someone is a new employee who did not miss the deadline (Dana or Bert).

This is **True**

(i)

$$\forall x(D(x) \leftrightarrow N(x))$$

Everyone missed the deadline if and only if they are a new employee. (Nima breaks it)

This is **False**

1.8.3

(d)

$$\exists x(P(x) \wedge C(x))$$

Negation: $\neg \exists x(P(x) \wedge C(x))$

Applying De Morgan's law: $\forall x \neg(P(x) \wedge C(x)) \equiv \forall x(\neg P(x) \vee \neg C(x))$

English: Every student showed up without a pencil or without a calculator.

(e)

$$\exists x(P(x) \vee C(x))$$

Negation: $\neg \exists x(P(x) \vee C(x))$

Applying De Morgan's law: $\forall x \neg(P(x) \vee C(x)) \equiv \forall x(\neg P(x) \wedge \neg C(x))$

English: Every student showed up without a pencil and without a calculator.

(f)

$$\forall x(P(x) \wedge C(x))$$

Negation: $\neg \forall x(P(x) \wedge C(x))$

Applying De Morgan's law: $\exists x \neg(P(x) \wedge C(x)) \equiv \exists x(\neg P(x) \vee \neg C(x))$

English: Some student showed up without a pencil or without a calculator.

1.8.4

(b)

$$\neg \forall x (\neg P(x) \rightarrow Q(x))$$

$$\exists x \neg (\neg P(x) \rightarrow Q(x)) \text{ Quantifier negation law}$$

$$\exists x \neg (\neg (\neg P(x)) \vee Q(x)) \text{ Implication law}$$

$$\exists x (\neg \neg (\neg P(x)) \wedge \neg Q(x)) \text{ De Morgan's law}$$

$$\exists x (\neg P(x) \wedge \neg Q(x)) \text{ Double negation law}$$

$$\text{Thus, } \neg \forall x (\neg P(x) \rightarrow Q(x)) \equiv \exists x (\neg P(x) \wedge \neg Q(x))$$

1.9.4

(d)

$$\exists x \forall y (P(x, y) \leftrightarrow P(y, x))$$

$\forall x \neg \forall y (P(x, y) \leftrightarrow P(y, x))$ Quantifier negation law

$\forall x \exists y \neg (P(x, y) \leftrightarrow P(y, x))$ Quantifier negation law

$\forall x \exists y \neg ((\neg P(x, y) \vee P(y, x)) \wedge (\neg P(y, x) \vee P(x, y)))$ Biconditional law

$\forall x \exists y ((\neg(\neg P(x, y) \vee P(y, x)) \wedge \neg(\neg P(y, x) \vee P(x, y)))$ De Morgan's law

$\forall x \exists y ((P(x, y) \wedge \neg P(y, x)) \wedge (\neg P(x, y) \wedge P(y, x)))$ Double negation and De Morgan's law

$\forall x \exists y ((P(x, y) \wedge \neg P(y, x)) \vee (\neg P(x, y) \wedge P(y, x)))$ Commutative law

1.10.3

(a)

P	a	b	c
a	T	T	T
b	F	F	F
c	T	T	F

For $\forall x \exists y P(x, y)$, each value of x needs at least one true entry in its row. For $x = b$ we have $P(b, a) = P(b, b) = P(b, c) = F$, so the statement is false.

For $\exists x \forall y P(x, y)$, one value of x needs every entry in its row true. Taking $x = a$, we have $P(a, a) = P(a, b) = P(a, c) = T$, so the statement is true.

Since one statement is false and the other is true for these values of P , the two are not logically equivalent.

1.10.4

(e)

$$\forall x \left((0 < x \wedge x < 1) \rightarrow \left(\frac{1}{x} > 1 \right) \right)$$

(f)

$$\neg \exists x \forall y (y < x)$$

$$\exists x \forall y (y \geq x)$$

(g)

$$\forall x (x \neq 0 \rightarrow \exists y (xy = 1))$$

1.10.7

(c)

$$\exists x(N(x) \wedge D(x))$$

(d)

$$\forall y(D(y) \rightarrow P(\text{Sam}, y))$$

(e)

$$\exists x(N(x) \wedge \forall y P(x, y))$$

(f)

$$\exists x(N(x) \wedge D(x)) \wedge \neg \exists y(N(y) \wedge D(y) \wedge y \neq x)$$

1.10.10

(c)

$$\forall x \exists y (y \neq \text{Math 101} \wedge T(x, y))$$

(d)

$$\exists x \forall y (y \neq \text{Math 101} \rightarrow T(x, y))$$

(f)

$$\exists y \exists z (T(\text{Sam}, y) \wedge T(\text{Sam}, z) \wedge y \neq z \wedge \forall w (T(\text{Sam}, w) \rightarrow (w = y \vee w = z)))$$

1.11.3

(b)

Form: $((p \vee q) \wedge \neg q) \rightarrow p$ (disjunctive syllogism)

p	q	$p \vee q$	$\neg q$	$((p \vee q) \wedge \neg q) \rightarrow p$
T	T	T	F	T
T	F	T	T	T
F	T	T	F	T
F	F	F	T	T

Every row gives T , so the argument is valid. When q is false the first premise forces p , and when q is true the second premise is false so the argument's hypotheses cannot both hold.

(c)

Form: $((p \rightarrow q) \wedge q) \rightarrow p$

p	q	$p \rightarrow q$	$((p \rightarrow q) \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

The third row gives F , so the argument is invalid (affirming the consequent). When $\sqrt{2}$ is rational and $2\sqrt{2}$ is still irrational, both premises are true while the conclusion is false, so the form has a counterexample.

1.11.4

(a)

Argument: 4 or 5 is prime. 5 is not prime. $\therefore$ 4 is prime.

Form: $((p \vee q) \wedge \neg q) \rightarrow p$

p	q	$p \vee q$	$\neg q$	$((p \vee q) \wedge \neg q) \rightarrow p$
T	T	T	F	T
T	F	T	T	T
F	T	T	F	T
F	F	F	T	T

Every row is T, so the argument is valid even though its conclusion p (4 is prime) is false. In this situation the second premise $\neg q$ (5 is not prime) is false, so the hypotheses cannot both be true and the false conclusion is never forced.

(b)

Argument: 5 is prime. If 4 is prime then 5 is prime. $\therefore$ 4 is not prime.

Form: $(q \wedge p \rightarrow q) \rightarrow \neg p$

p	q	q	$p \rightarrow q$	$q \wedge (p \rightarrow q)$	$\neg p$
T	T	T	T	T	F
T	F	F	T	F	F
F	T	T	T	T	T
F	F	F	T	F	T

In the first row both hypotheses are T while the conclusion $\neg p$ is F, so the form is invalid. Here p is F and q is T, so both hypotheses are true and the conclusion 4 is not prime is true, giving an invalid argument with a true conclusion.